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Comparative Evaluation of Deep Learning Architectures for Glaucoma Screening: From Internal Accuracy to Cross-Domain

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Abstract

Reliable cup-to-disc ratio estimation is essential for automated glaucoma screening, yet segmentation robustness remains challenging under illumination and anatomical variability. This study compares three deep learning architectures—UPerNet, MAnet, and Dense Prediction Transformer (DPT)—to identify the most reliable model for clinical deployment. Using 5-fold cross-validation and Bayesian optimization, models were evaluated on region, boundary, and clinical metrics. UPerNet achieved the highest segmentation overlap (Cup Dice=0.8750; Disc Dice=0.8941) and the lowest clinical error (CDR MAE=0.0849), outperforming both DPT and MAnet. Furthermore, zero-shot external validation on the REFUGE dataset confirmed UPerNet's strong generalization capabilities (Cup Dice=0.8791; Disc Dice=0.8737; CDR MAE=0.0752). We conclude that carefully tuned multi-scale CNNs, particularly UPerNet, provide the most reliable segmentations for automated glaucoma assessment.

Keywords: Glaucoma; Deep Learning; Semantic Segmentation; Fundus Images; Cup-to-Disc Ratio.

1. Introduction

Glaucoma is one of the leading causes of irreversible blindness worldwide [1], affecting more than 76 million people globally, with projections exceeding 110 million cases by 2040. The disease causes progressive damage to retinal ganglion cells and the optic nerve. Because early stages are typically asymptomatic, significant and irreversible vision loss often occurs before diagnosis, making early detection clinically essential.

Fundus imaging is widely used for glaucoma screening. Accurate segmentation of the optic disc (OD) and optic cup (OC) enables computation of the cup-to-disc ratio (CDR), an important biomarker for glaucoma diagnosis and monitoring. However, automatic segmentation remains challenging due to anatomical variability, vessel occlusion, low contrast, and uneven illumination.

Deep learning, particularly convolutional neural networks (CNNs), has significantly improved medical image segmentation. More recent approaches incorporate attention mechanisms and global context modeling to better capture long-range dependencies and improve boundary delineation. Nevertheless, few studies provide controlled comparisons between CNN-based and transformer-based architectures under identical training and evaluation settings.

In this work, we compare three segmentation paradigms: feature pyramid-based modeling, transformer-based dense prediction, and multi-scale attention mechanisms. Specifically, UPerNet, Dense Prediction Transformer (DPT), and MAnet are evaluated under the same experimental protocol. The objective is to assess segmentation performance and its impact on clinically relevant CDR estimation. Additionally, the robustness of the best-performing model is evaluated through external zero-shot validation on the public REFUGE dataset.

2. State of the Art

Recent approaches for optic disc and cup segmentation increasingly combine convolutional architectures with attention mechanisms and optimization strategies to improve robustness and segmentation accuracy under varying imaging conditions.

DB-SegNet [2] extends the conventional SegNet architecture by integrating a Dilated Atrous Context Module and a Bidirectional Feature Calibration Unit to enhance multi-scale feature extraction and boundary refinement. The framework also incorporates optimization techniques such as Bitterling Fish Optimization for feature selection and Honey Badger Optimization for hyperparameter tuning, together with a Multi-Scale Attention Transformer. Reported results include Dice scores of 99.2% for the optic disc and 98% for the optic cup. However, the pipeline is highly complex, which may limit reproducibility and computational efficiency.

Another study proposed an Attention U-Net enhanced with residual connections and the CBAM attention module [3]. The architecture aims to improve feature representation and reduce gradient vanishing effects. Although the reported sensitivity and specificity are competitive, the absence of cross-dataset validation limits conclusions regarding generalization to unseen clinical data.

A third approach combines SegFormer for segmentation with a Swin Transformer for classification [4]. The method achieves high reported performance while reducing computational complexity. Nevertheless, evaluation was conducted using internal dataset splits, which may not accurately reflect real-world clinical performance.

Overall, recent methods report strong segmentation results, but several limitations remain. Comparisons across datasets are inconsistent, reproducibility is often limited, and clinical validation remains insufficient. In particular, few studies evaluate segmentation performance together with clinically relevant indicators such as the cup-to-disc ratio.

3. Methodology

This study addresses multi-class semantic segmentation of the optic disc (OD) and optic cup (OC) in retinal fundus images. The proposed pipeline consists of data preprocessing, dataset construction, model training, hyperparameter optimization, cross-validation, ensemble inference, and evaluation. Each architecture was trained and optimized in a separate but identically configured experimental pipeline, ensuring that preprocessing, hyperparameter search space, loss function, and cross-validation strategy remained strictly consistent across all models.

3.1. Dataset

The dataset employed in this study consists of color fundus images (retinographies) accompanied by segmentation masks manually annotated by clinical experts. These images present considerable variability in illumination, contrast, and anatomical structure, reflecting the heterogeneity typical of real-world clinical data. The segmentation task uses three semantic classes: Background (Class 0), Optic Cup (Class 1), and Optic Disc (Class 2). These three classes are required to determine the cup-to-disc ratio (CDR). The original masks were provided in grayscale and were processed using simple intensity thresholding. Pixels with intensity $I \geq 0.75$ were assigned to Class 1 (Cup);

intensities in the range $0.25 \leq I < 0.75$ were mapped to Class 2 (Disc); and all pixels with $I < 0.25$ were assigned to Background (Class 0). In addition to the primary dataset used for training and internal cross-validation, the public REFUGE (Retinal Fundus Glaucoma Challenge) dataset was utilized exclusively for external zero-shot validation. Unlike the primary dataset, which consists of localized crops around the optic disc, the REFUGE dataset provides full fundus images with different illumination profiles and camera sensors. This structural and visual difference introduces a significant Field of View (FoV) discrepancy and domain shift, making it an ideal independent benchmark to evaluate the model's true generalization capabilities and robustness in real-world clinical scena

3.2. Preprocessing, Normalization and Data Augmentation

A preprocessing pipeline was implemented to ensure compatibility with standard deep learning architectures and to promote numerical stability during training:

- All images and their corresponding masks were resized to a fixed resolution of 224 × 224 pixels. Linear interpolation was applied to the images, while nearest-neighbor interpolation was used for the masks to avoid introducing invalid label values.

- Pixel intensities were normalized to the range [0, 1] by scaling each value by a factor of 1/255. This normalization maintains a stable input distribution, which is particularly important when using pre-trained encoders, as all models evaluated in this work rely on ImageNet [5] pre-training.

- Data augmentation was applied during training to improve generalization. The following transformations were used: random horizontal flipping ($p = 0.5$), random vertical flipping ($p = 0.5$), random rotations up to ± 15°, and random brightness and contrast adjustments within a limited range. All spatial transformations were applied consistently to both images and their corresponding masks.

3.3. Model Architecture

Three deep learning architectures for semantic segmentation were compared in this study, with the objective of determining which approach performs most effectively for fundus image analysis. All three models employ encoders pre-trained on ImageNet [5], providing a robust foundation for feature extraction.

- MAnet (Multi-scale Attention Network) is built on a ResNet34 [6] backbone, originally proposed for liver and tumor segmentation [7]. Its core innovation is a multi-scale attention mechanism that fuses features across different resolutions. The decoder comprises five stages, progressively reducing the channel dimension from 256 to 16. A Position Attention Block (PAB) is integrated into the design to enhance the model's ability to capture global spatial dependencies.

- DPT (Dense Prediction Transformer) [8] adopts a fundamentally different strategy by employing a Vision Transformer backbone, specifically the vit_base_patch16_224 variant. In this architecture, convolutional downsampling is replaced by patch embeddings and global self-attention mechanisms. This design preserves high-resolution representations throughout the network while effectively capturing long-range dependencies. Deep conventional CNNs tend to lose contextual information as network depth increases; DPT is explicitly designed to mitigate this limitation.

- UPerNet (Unified Perceptual Parsing Network) [9] returns to a ResNet34 [6] encoder but integrates it within a Feature Pyramid Network (FPN) framework to aggregate multi-scale contextual information. The decoder operates with 256 channels. This configuration balances high-level semantic features with fine spatial details, a desirable property for delineating cup boundaries while maintaining awareness of global fundus structures.

All three architectures share the same final stage: a convolutional head that maps latent features to the three semantic classes, followed by a Softmax activation function. This produces pixel-wise probability maps for each input image.

3.4. Hyperparameter Optimization

Hyperparameter optimization was performed using the Optuna framework [10] with a Tree-structured Parzen Estimator (TPE) sampler [11]. The objective function minimized the validation loss on a dedicated holdout subset separated from the main training pool to prevent data leakage. The search space included the learning rate, sampled on a logarithmic scale in the range $[1 \times 10^{-5}, 1 \times 10^{-3}]$; the batch size, selected from {4, 8} due to GPU memory constraints; and the Dice loss weight α, sampled in the interval [0.3, 0.7]. For each trial, models were trained for 3 epochs to provide a computationally efficient approximation of convergence behavior. Validation performance was evaluated using a composite loss function combining Dice Loss and Weighted Cross-Entropy Loss, which also served as the optimization objective. This optimization procedure was executed separately for each architecture using the same search space configuration. The resulting hyperparameters were then used for full training under a 5-fold cross-validation scheme to ensure robust evaluation.

3.5. Loss Function

A composite loss function combining Weighted Cross- Entropy and Dice Loss [13] was employed for training the MAnet, DPT, and UPerNet models. This formulation addresses both pixel-wise classification accuracy and the severe class imbalance inherent in fundus image segmentation. The Weighted Cross-Entropy Loss is defined as:

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c y_{i,c} \log(p_{i,c}) \tag{1}$$

where N is the number of pixels, C is the number of classes, y_i,c is the one-hot encoded ground-truth label for pixel i, and p_i,c is the predicted probability for class c. The weights w_c are computed based on the inverse class frequency in the training set to mitigate class imbalance. The Dice Loss is defined as:

$$L_{Dice} = 1 - \frac{2\sum_{c=1}^{C}\sum_{i=1}^{N} p_{i,c} y_{i,c} + \varepsilon}{\sum_{c=1}^{C}\sum_{i=1}^{N} p_{i,c} + \sum_{c=1}^{C}\sum_{i=1}^{N} y_{i,c} + \varepsilon} \tag{2}$$

where ε is a smoothing term to avoid division by zero. The final loss is given by:

$$L_{total} = \alpha L_{Dice} + (1 - \alpha) L_{CE} \tag{3}$$

where $\alpha \in [0, 1]$ is a weighting factor optimized empirically using Optuna.

3.6. Training Protocol and Ensemble Inference

Cross-Validation and Data Splitting: a 5-fold cross-validation (5-fold CV) strategy was implemented to ensure robust performance estimation [12]. The dataset was partitioned into five disjoint subsets. In each iteration, four folds were used for training while the remaining fold was used for validation. This procedure ensures that all samples are used for both training and validation, reducing bias and variance associated with a single data split. In training execution, each model was trained for 50 epochs per fold. The validation loss was computed at the end of every epoch. A best model saving strategy was adopted, where the model weights corresponding to the lowest validation loss were retained. This prevents performance degradation due to overfitting in later epochs and ensures that the most generalizable model is used for inference. An ensemble strategy was employed to improve prediction. Instead of relying on a single model, the predictions from the five models obtained during cross-validation were aggregated. For each pixel, the Softmax probability maps were averaged. The final class label was assigned using the argmax operator.

This approach reduces prediction variance and improves segmentation consistency, particularly along ambiguous boundaries such as the optic cup and disc interface.

3.7. Evaluation Metrics

The performance of the segmentation models was quantitatively evaluated using three categories of metrics. First, region-based metrics including Intersection over Union (IoU) and the Dice Coefficient were used to measure the global overlap between predicted masks and ground truth. Second, boundary-based metrics, specifically the Hausdorff Distance (HD) and Boundary IoU (BIoU), were employed to assess the precision of the predicted contours. Finally, the Cup-to-Disc Ratio (CDR) Error was calculated as a clinical validation metric, representing the absolute difference between the manual and automated CDR measurements, which is critical for glaucoma screening. Boundary IoU was computed using narrow dilated contour bands (1-pixel dilation), resulting in a stricter and numerically lower overlap measure than conventional region-based IoU metrics.

3.8. External Validation and Oracle Localization

To assess the generalizability of the proposed pipeline against real-world domain shifts (e.g., varying sensors and illumination), the best-performing model was subjected to a zero-shot external validation using the public REFUGE2 test set. A significant challenge in cross-dataset fundus evaluation is the Field of View (FoV) discrepancy: our primary dataset consists of Region of Interest (ROI) crops centered on the optic disc, whereas REFUGE provides full fundus images. To isolate the model's pure segmentation capability from localization failures caused by this scale mismatch, we simulated a two-stage clinical pipeline using an "Oracle Localization" approach. Prior to inference, bounding boxes were extracted utilizing the ground-truth annotations to consistently crop the optic disc region. These cropped images were then normalized and resized to 224 x 224 pixels, matching the scale of the training distribution, allowing for a fair evaluation of the model's contour delineation robustness under domain shift.

4. Quantitate Results

This section presents the experimental results obtained for the automated segmentation of the optic disc (Class 2) and optic cup (Class 1) in retinal fundus images for glaucoma assessment.

4.1. Hypermeter Optimization

The optimal hyperparameter configuration shows that Transformer-based architectures, particularly DPT, require substantially lower learning rates to achieve stable convergence compared with CNN-based models such as UPerNet and MAnet, as summarized in Table 1. The batch size remains constant across all experiments (8), ensuring comparability across architectures. In contrast, the Dice loss weighting factor varies moderately depending on the model, reflecting differences in optimization sensitivity and segmentation behavior. DPT exhibits the lowest learning rate (1.7×10^{-5}), whereas UPerNet and MAnet converge with higher learning rates of 6.56×10^{-4} and 3.55×10^{-4}, respectively.

4.2. Region-based Segmentation Performance

Region-based performance was evaluated using IoU and Dice, computed on the test set for the optic cup (Class 1) and optic disc (Class 2). These metrics quantify global overlap and are the most reported measures in medical segmentation. Table 1 reports the mean scores for each model.

Table 1. Mean IoU and Dice on the test set.

Model	Cup (Class 1)		Disc (Class 2)	
	IoU	Dice	IoU	Dice
MAnet	0.7805	0.8690	0.8091	0.8921
UPerNet	0.7893	0.8750	0.8119	0.8941
DPT	0.7853	0.8729	0.8027	0.8883

4.3. Boundary Quality

Clinical usability depends not only on area overlap but also on contour accuracy, we additionally assessed boundary quality using the Hausdorff Distance (HD) and Boundary IoU (BIoU). HD captures worst-case contour deviations, while BIoU focuses on overlap restricted to boundary bands, making it more sensitive to edge localization errors. The mean boundary results are shown in Table 2.

Table 2. Boundary metrics

Model	Cup (Class 1)		Disc (Class 2)	
	HD	BIoU	HD	BIoU
MAnet	7.4164	0.0540	8.4643	0.0592
UPerNet	7.1965	0.0564	8.5064	0.0601
DPT	7.3858	0.0500	8.8833	0.0536

4.4. Clinical Metric: CDR Error

To connect segmentation quality with clinical decision support, we computed the cup-to-disc ratio (CDR) from predicted masks and measured the mean absolute error (MAE) relative to the ground truth CDR. This metric directly reflects the reliability of downstream glaucoma screening indicators. Table 3 reports the mean CDR MAE for each architecture.

Table 3 CDR (MAE)

Model	CDR MAE
DTP	0.0954
UPerNet	0.0849
MAnet	0.1007

4.5 External Validation on REFUGE Dataset

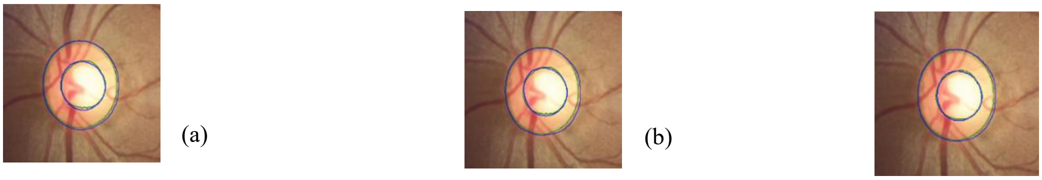
Following the internal evaluation, the UPerNet ensemble was selected as the optimal architecture and applied directly to the REFUGE dataset for zero-shot external validation. Under the Oracle Localization setup, the model maintained robust segmentation accuracy across domains without any further fine-tuning. As detailed in Table X, UPerNet achieved consistent region and boundary metrics for both the optic cup and disc. More importantly, the clinical error improved slightly relative to the internal test set, yielding a Cup-to-Disc Ratio Mean Absolute Error (CDR MAE) of 0.0752. These findings confirm that UPerNet's feature representations are highly resilient to variations in contrast and illumination, provided the anatomical scale remains consistent. Table 4 presents the performance:

Table 4. Zero-shot external validation performance of UPerNet on the REFUGE dataset

Structure	IoU	Dice	HD	BIoU
Cup (Class 1)	0.7879	0.8791	5.90	0.0501
Disc (Class 2)	0.7788	0.8737	7.10	0.0409

5. Qualitive Results

A representative qualitative comparison of the three architectures is presented in Fig. 1, showing the predicted contours of the optic cup and disc alongside the corresponding ground-truth. For this specific example, the DPT model achieved a Cup IoU of 0.923 (Dice: 0.960) and a Disc IoU of 0.886 (Dice: 0.939). The MANet model showed slight improvements in the cup segmentation with a Cup IoU of 0.929 (Dice: 0.963) and a Disc IoU of 0.885 (Dice: 0.939). Finally, the UPerNet model yielded the highest overlap for this case, reaching a Cup IoU of 0.943 (Dice: 0.971) and a Disc IoU of 0.897 (Dice: 0.946).



(a) (b) (c)

Figure 1.Qualitative evaluation of optic disc and optic cup segmentation. (a) DPT prediction; (b) MAnet prediction; (c) UPerNet prediction. Green contours denote the expert ground truth, while blue contours represent the model predictions.

6. Discussion

The results show that, under an identical pipeline, the three architectures achieve broadly similar region-based performance but differ more clearly in boundary quality and clinical reliability. Visual inspection indicates that illumination variability is a major source of error, with low contrast, overexposure, vessel occlusions, and shading primarily affecting optic cup delineation. These factors have limited influence on Dice/IoU but substantially impact boundary metrics and derived CDR estimates.

UPerNet achieves the best overall performance, with the highest Dice scores for both cup (0.8750) and disc (0.8941), alongside the strongest boundary consistency, reflected in the best BIoU values (cup 0.0564; disc 0.0601) and the lowest cup Hausdorff distance (7.1965). DPT, despite competitive overlap performance, shows weaker boundary alignment (cup BIoU 0.0500; disc 0.0536) and a higher disc Hausdorff distance (8.8833), indicating smoother but less anatomically precise contours. MAnet remains close in Dice yet exhibits higher CDR error (0.1007), suggesting systematic bias not captured by overlap-based metrics. From a clinical perspective, these differences directly propagate to CDR estimation. UPerNet achieves the lowest mean absolute error (0.0849), followed by DPT (0.0954) and MAnet (0.1007), confirming that boundary fidelity has a stronger impact on diagnostic reliability than global overlap. The zero-shot evaluation on the REFUGE dataset further supports these findings: direct transfer initially fails due to field-of-view mismatch, but after Oracle-based localization, UPerNet preserves strong segmentation performance and maintains a clinically acceptable CDR error (0.0752), demonstrating robustness under domain shift.

Overall, UPerNet provides the most consistent trade-off between overlap accuracy, boundary precision, and clinical stability. Transformer-based DPT remains competitive in region-level metrics but shows reduced robustness in boundary-sensitive and cross-domain settings, consistent with its higher sensitivity during optimization. Error analysis further shows that performance is not uniformly distributed across anatomical structures. The optic cup is more sensitive to intensity variation and vessel crossings than the optic disc, producing higher variance in cup-related metrics across all models. This imbalance directly affects CDR stability, since small cup segmentation errors propagate nonlinearly into ratio estimation. Failure cases are also concentrated in images with extreme illumination conditions, indicating limited robustness to photometric shifts typical of real-world screening data. These observations suggest that clinical reliability depends less on global feature extraction capacity and more on precise boundary modeling under degraded imaging conditions.

7. Conclusion

This study presented a rigorous comparative analysis of three advanced deep learning architectures—UPerNet, MAnet, and DPT—for optic disc and cup segmentation in retinal fundus images. By employing a standardized experimental pipeline with Bayesian hyperparameter optimization and 5-fold cross-validation, we ensured a robust and fair evaluation encompassing region-based overlap, boundary fidelity, and clinical efficacy.

Our findings demonstrate that multi-scale CNN-based architectures, specifically UPerNet with feature pyramid aggregation, provide the most reliable performance for this task. UPerNet not only achieved the highest region overlap but, crucially, delivered superior contour alignment and the lowest Cup-to-Disc Ratio (CDR) error. While the Transformer-based DPT and the attention-driven MAnet yielded competitive regional overlap, their higher boundary deviations and clinical errors underscore a critical takeaway: standard segmentation metrics (like Dice and IoU) alone are insufficient to guarantee clinical reliability in automated glaucoma screening. Furthermore, external validation on the public REFUGE dataset confirmed the model's exceptional robustness to domain shifts when spatial scale is controlled, maintaining state-of-the-art regional overlap and clinical accuracy.

Ultimately, accurate boundary delineation is paramount for calculating dependable clinical indicators. Future work will focus on validating these findings across diverse, multi-centric datasets to assess cross-domain generalization, and on exploring lightweight architectural adaptations to facilitate real-time deployment in resource-constrained clinical environments.

8. References

[1] Pannu Rabia, Zubair Muhammad, Owais Muhammad, Hassan Shoaib, Umair Muhammad, Usman Syed Muhammad, Albashrawi Mousa Ahmed and Hussain Irfan (2025) “Enhanced glaucoma classification through advanced segmentation by integrating cup-to-disc ratio and neuro-retinal rim features", Computerized Medical Imaging and Graphics 123: 102-559.

[2] Prakash J. and Kumar B.Vinoth (2025) “DB-SegNet: optimized framework for glaucoma detection and optic structure segmentation from retinal fundus images”, Scientific Reports 15 (1): 39843.

[3] Vanaja C.B. and Prakasam P. (2025) ``Convolutional block attention gate-based U-Net framework for microaneurysm segmentation using retinal fundus images'', BMC Medical Imaging 25 (1).

[4] Tofa F.S., Hassan M., Mahmud U. and Yusuf L. (2025) ``A new glaucoma detection method using a Swin Transformer and image segmentation'', Engineering Proceedings 87 (1): 113

[5] Deng J., Dong W., Socher R., Li L-J, Li K. and Fei-Fei L. (2009) ``ImageNet: A large-scale hierarchical image database'' iIEEE Conf. Computer Vision and Pattern Recognition (CVPR) 248-255.

[6] He K., Zhang X., Ren S. and Sun J. (2016) ``Deep residual learning for image recognition'' iIEEE Conf. Computer Vision and Pattern Recognition (CVPR) 770-778.

[7] Fan T., Wang G., Li Y. and Wang H. (2020) ``MA-Net: A multi-scale attention network for liver and tumor segmentation,'' IEEE Access 8:179656—179665

[8] Ranftl R., Bochkovskiy A. and Koltun V. (2021) ``Vision transformers for dense prediction”, IEEE/CVF Int. Conf. Computer Vision (ICCV) 12179--12188.

[9] Xiao T., Liu Y., Zhou B, Jiang Y. and Sun J. (2018) ``Unified perceptual parsing for scene understanding,'' Proc. European Conf. Computer Vision (ECCV) 418-434.

[10] Akiba T., Sano S., Yanase T., Ohta T. and Koyama M. (2019) ``Optuna: A next-generation hyperparameter optimization framework'', Conference: the 25th ACM SIGKDD International Conference Anchorage, AK, USA 2623-2631.

[11] Bergstra J., Bardenet R., Bengio Y. and Kégl B. (2011) ``Algorithms for hyper-parameter optimization'', Advances in Neural Information Processing Systems (NeurIPS) 24: 2546-2554.

[12] Wilimitis D. and Walsh C. G. (2023) ``Practical considerations and applied examples of cross-validation for model development and evaluation in health care: Tutorial'', JMIR AI 2: 49023.